CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building LLM-powered applications and agents from prototype to production, with RAG, tool integrations and evaluation that shows whether a solution actually works. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by AI consulting work with business stakeholders on where agentic AI fits their workflows.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Owned the knowledge-extraction data pipelines end-to-end as sole developer, from data ingestion and chunking to prompt engineering, context handling and indexing, reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Built LLM agents in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documents, and evaluated them against an SAP-system-derived ground truth test set, raising precision/recall from 33%/45% to 93%/90%.
- Built a deterministic, code-based pipeline instead of an LLM for a second, high-volume data stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed a Model Context Protocol (MCP) service that exposes the knowledge graph as a tool integration for agents, with millisecond-latency retrieval, integrated into Claude Code.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, FastAPI, Postgres, MongoDB, Neo4j, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Advised business stakeholders as an internal consultant: mapped their decision-making workflows, identified where agentic AI could support them and translated their requirements into a solution approach.
- Built and deployed an agentic AI application with LangGraph that routes requests across vector-database retrieval, general inference and summarization.
- Built the Python/FastAPI backend (APIs tested with Postman) and a React frontend, containerized with Docker and deployed to AWS with CDK; version control in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated evaluation of RAG retrieval and generation with RAGAS and PytrecEval against chemist-provided ground truth, running as an evaluation pipeline on AWS (Lambda, SQS, DynamoDB) built with CDK for PrismaChat.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking strategies, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models on time-series process data with engineered lag and rolling-window features, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: Agent Eval Harness | Python | LangGraph | GitHub Actions

- Evaluation harness for a tool-calling LangGraph agent over a mocked API: golden test set for evaluations (regression, edge and adversarial cases), deterministic automated checks first (JSON schema, tool-call payloads, API status), then LLM-as-a-judge.
- Repeated runs with pass rates, prompt-injection and tool-loop tests, traces and a pass-rate view in Arize Phoenix, run in GitHub Actions on every change.

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA Fine-Tuning

- Fine-tuned an Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

TECHNICAL SKILLSET

- **LLM Applications & Agents:** LLM APIs, RAG, agents and tool integrations (LangGraph, LangChain, MCP), prompts and context handling, LiteLLM, Hugging Face Transformers, QLoRA fine-tuning
- **Evaluation:** test sets against ground truth, automated checks, RAGAS, PytrecEval, precision/recall
- **Software Engineering:** Python, FastAPI, REST APIs, testing (Postman), version control (Git, GitLab CI/CD), React (beginner)
- **Data & Cloud:** SQL, Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search), Pandas, AWS (Lambda, SQS, DynamoDB, EC2, S3, CDK), Azure, Docker, Kubernetes
- **Ways of working:** stakeholder requirements and advisory, AI-assisted development with Claude Code, Agile

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)